STEPS TO CREATE BASIC PYTHON FILE

Step-1:Create a Project Folder

mkdir python-docker-app

cd python-docker-app

Step 2: Create Python Script

Create a file app.py inside your folder:

app.py

print("Hello from Python inside Docker!")

Step 3: Create requirements.txt

Create a text file inside your folder:

requirements.txt

```
flask
numpy
```

Step 3: Create Dockerfile

In the same folder, create a file named Dockerfile (no extension):

Use official Python image

FROM python:3.10-slim

Set working directory in container

WORKDIR /app

Copy project files into container

COPY . .

Install dependencies (if you have requirements.txt)

RUN if [-f requirements.txt]; then pip install -r requirements.txt; fi

Run the Python script

CMD ["python", "app.py"]

Step 5: Build Docker Image

docker build -t python-docker-app .

Step 6: Run the Docker Container

docker run --name python-container python-docker-app

Optional: Run Interactively

If your Python script requires input:

docker run -it python-docker-app

Tasks to do

1.Create a python script inside Docker to generate a random list of 10 numbers

2..Create a python script inside Docker to compute max, min, and average of numbers(Take input from the user)

3..Create a python script inside Docker to display a funny dialogue when you enter the name of the actor

STEPS TO CREATE BASIC JAVA FILE

Step-1:Create a Project Folder

mkdir java-app

cd java-app

Step 2: Create Python Script

Create a file App.java inside your folder:

```
# App.java

import java.util.Scanner;

import java.io.*;


class App {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a value: ");

        int a = sc.nextInt();
```

```
        System.out.print("Enter b value: ");
        int b = sc.nextInt();
        int c = a + b;
        System.out.println("The sum of two values is: " + c);
        sc.close();
    }
}
```

Step 3: Create Dockerfile

In the same folder, create a file named Dockerfile (no extension):

```
# Use official Python image
# Use official OpenJDK image
FROM openjdk:17-jdk-slim
# Set working directory
WORKDIR /app
# Copy Java source file
COPY App.java /app
# Compile Java program
RUN javac App.java
# Run Java program
CMD ["java", "App"]
```

Step 4: Build Docker Image

```
docker build -t java-app .
```

Step 6: Run the Docker Container

```
docker run java-app
```

Optional: Run Interactively

If your java program requires input:

docker run -it java-app

Tasks to do:

1.Write a java program inside Docker to create a parametarized constructor to apply addition and multiplication on the input values.

2.Write a java program inside Docker to create a single inheritance.

3.Write a java program inside Docker to create a palindrome.